Exponentials & Logarithms

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Log laws: product and quotient**

Write $\log_2(8x)$ as a sum of logarithms.

$\log_2(8x) = \log_2 8 + \log_2 x = 3 + \log_2 x$.

2 **[Foundational]** • **Target: Log laws: product and quotient**

Write $\log_3\left(\frac{x}{9}\right)$ as a difference of logarithms.

$\log_3\left(\frac{x}{9}\right) = \log_3 x - \log_3 9 = \log_3 x - 2$.

3 **[Foundational]** • **Target: Log laws: product and quotient**

Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.

$\log_5(5 \times 25) = \log_5 125 = 3$ (since $5^3 = 125$).

4 **[Foundational]** • **Target: Log laws: the power rule**

Write $\log(x^5)$ using the power rule.

$\log(x^5) = 5\log x$.

5 **[Foundational]** • **Target: Log laws: the power rule**

Evaluate $\log_2(4^3)$ using the power rule.

$\log_2(4^3) = 3\log_2 4 = 3(2) = 6$.

6 **[Foundational]** • **Target: Log laws: the power rule**

Write $2\log_3 x + \log_3 y$ as a single logarithm.

$2\log_3 x + \log_3 y = \log_3(x^2 y)$.

7 **[Foundational]** • **Target: Expanding and condensing logarithmic expressions**

Expand $\log\left(\frac{x^3 y}{z^2}\right)$ fully.

$3\log x + \log y - 2\log z$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Expanding and condensing logarithmic expressions

Condense $3\ln x - 2\ln y$ into a single logarithm.

$\ln\left(\frac{x^3}{y^2}\right)$.

9 [Intermediate] • Target: Solving exponential equations using logarithms

Solve $2^x = 10$ for x, rounded to three decimal places.

$x = \frac{\log 10}{\log 2} \approx \frac{1}{0.301} \approx 3.322$.

10 [Intermediate] • Target: Natural logarithms and e

Solve $\ln(x) = 2$ for x, in terms of e.

$x = e^2$.

11 [Intermediate] • Target: Exponential growth and decay models

The population of a town is modeled by $P(t) = 5000e^{0.03t}$, where t is in years. Find $P(10)$, rounded to the nearest whole number.

$P(10) = 5000e^{0.3} = 5000(1.34986) \approx 6749$.

12 [Intermediate] • Target: Exponential growth and decay models

A radioactive substance decays according to $A(t) = 200e^{-0.05t}$ (grams). Find $A(20)$, rounded to one decimal place.

$A(20) = 200e^{-1} = 200(0.36788) \approx 73.6$ grams.

13 [Intermediate] • Target: Exponential growth and decay models

Determine whether $N(t) = 50e^{0.02t}$ models growth or decay, and justify your answer.

Growth — the exponent's coefficient 0.02 is positive, so N increases as t increases.

14 [Intermediate] • Target: Word problems: exponential models in context

This question has two parts. An investment of 2000 dollars grows according to $A(t) = 2000e^{0.04t}$, where t is in years.

a Find the value of the investment after 5 years, rounded to the nearest dollar.
$A(5) = 2000e^{0.2} = 2000(1.2214) \approx 2443$ dollars.

b Find how many years it takes for the investment to double, rounded to one decimal place (using $\ln 2 \approx 0.693$).
$2 = e^{0.04t} \Rightarrow 0.04t = \ln 2 \approx 0.693 \Rightarrow t \approx 17.3$ years.

(a) $A(5) = 2000e^{0.2} = 2000(1.2214) \approx 2443$ dollars.; (b) $2 = e^{0.04t} \Rightarrow 0.04t = \ln 2 \approx 0.693 \Rightarrow t \approx 17.3$ years.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Word problems: exponential models in context** • **Cross-Lesson: 6.01 Limits & Derivatives**

This question has two parts. A cup of coffee cools according to $T(t) = 20 + 60e^{-0.1t}$ (°C), where t is minutes after brewing and 20°C is room temperature.

a Find the temperature after 15 minutes, rounded to one decimal place (using $e^{-1.5} \approx 0.2231$).

$T(15) = 20 + 60e^{-1.5} = 20 + 60(0.2231) \approx 33.4$°C.

b Find how many minutes it takes for the temperature to drop to 35°C, rounded to one decimal place (using $\ln 4 \approx 1.386$).

$35 = 20 + 60e^{-0.1t} \Rightarrow e^{-0.1t} = 0.25 \Rightarrow -0.1t = -\ln 4 \approx -1.386 \Rightarrow t \approx 13.9$ minutes.

(a) $T(15) = 20 + 60e^{-1.5} = 20 + 60(0.2231) \approx 33.4$°C.; (b) $35 = 20 + 60e^{-0.1t} \Rightarrow e^{-0.1t} = 0.25 \Rightarrow -0.1t = -\ln 4 \approx -1.386 \Rightarrow t \approx 13.9$ minutes.

16 **[Challenging]** • **Target: MCQ — Expanding and condensing logarithmic expressions** • **Cross-Lesson: 7.01 Integration & Applications**

Evaluate the exact expression: $5^{\log_5(12)} + \log_7(1)$

Step 1: Apply inverse property $b^{\log_b x} = x$: $5^{\log_5(12)} = 12$. Step 2: Apply zero property $\log_b(1) = 0$: $\log_7(1) = 0$. Step 3: Sum the terms: $12 + 0 = \mathbf{12}$.

17 **[Challenging]** • **Target: MCQ — Solving exponential equations (matching bases)** • **Cross-Lesson: 9.01 Statistics & Probability**

Geometric sequences are used to model population growth. If the current population of a city is 250,000 and it is projected to double every 15 years, what will be the estimated population after 60 years?

This problem represents a geometric sequence where the first term (a) is the current population, and the common ratio (r) is 2 (since the population doubles). The population doubles every 15 years. We want to find the population after 60 years. The number of doubling periods is: $\frac{60 \text{ years}}{15 \text{ years/doubling}} = 4 \text{ doublings}$ The formula for a geometric sequence is $a_n = a \cdot r^{(n-1)}$, where a is the first term, r is the common ratio, and n is the term number. In this.

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18 [Challenging] • Target: MCQ — Solving exponential equations using logarithms • Cross-Lesson: 1.01 Algebra & Equations

If $\log_3(x^2 - 2x + 1) = 2$, find the values of x that satisfy this equation.

To solve the equation $\log_3(x^2 - 2x + 1) = 2$, we convert it to exponential form. Recall that $\log_b a = c$ is equivalent to $b^c = a$. Therefore, $3^2 = x^2 - 2x + 1$. Simplifying the equation: $9 = x^2 - 2x + 1$ We rearrange the constants to one side to form a quadratic equation: $x^2 - 2x + 1 - 9 = 0$ $x^2 - 2x - 8 = 0$ We solve the quadratic equation by factoring or using the quadratic formula. By factoring, we look for two numbers that multiply to -8 and add to -2. These numbers are -4 and 2. Thus, the equ.

19 [Challenging] • Target: MCQ — Solving logarithmic equations • Cross-Lesson: 2.01 Functions & Transformations

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

Take the natural logarithm of both sides: $\ln(3^{2x+1}) = \ln(5^{x-2})$. Apply the power law of logarithms: $(2x + 1)\ln 3 = (x - 2)\ln 5$. Distribute: $2x\ln 3 + \ln 3 = x\ln 5 - 2\ln 5$. Collect all terms containing x on one side: $2x\ln 3 - x\ln 5 = -\ln 3 - 2\ln 5 \Rightarrow x(2\ln 3 - \ln 5) = -(\ln 3 + 2\ln 5)$. Simplify logarithmic coefficients: $2\ln 3 = \ln 9$ and $2\ln 5 = \ln 25$: $x(\ln 9 - \ln 5) = -(\ln 3 + \ln 25) \Rightarrow x\ln\left(\frac{9}{5}\right) = -\ln(75)$. Solve for x.

20 [Challenging] • Target: MCQ — Natural logarithms and e • Cross-Lesson: 3.01 Sequences & Series

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

Combine the logarithms on the left-hand side using the product rule: $\log_3[x(x + 6)] = 3$. Convert to exponential form: $x(x + 6) = 3^3 = 27 \Rightarrow x^2 + 6x - 27 = 0$. Factor the quadratic: $(x + 9)(x - 3) = 0 \Rightarrow x = -9$ or $x = 3$. Check the domain restrictions in the original equation:For $x = -9$: $\log_3(-9)$ is undefined, so $x = -9$ is an extraneous solution.For $x = 3$: $\log_3(9) + \log_3(3) = 2 + 1 = 3$. Valid ✓.Therefore, the unique solution is $x = 3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Exponential growth and decay models

Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.

a. Decay ✓

b. Neither

c. Growth

d. Cannot be determined

The exponent's coefficient is negative: decay.

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22 [Foundational] • Target: MCQ — Exponential growth and decay models

The graph shows an exponential function. Based on the graph, does it represent growth or decay?

a. Neither

b. **Growth ✓**

c. Cannot be determined

d. Decay

The curve rises as x increases: growth.

23 [Foundational] • Target: MCQ — Word problems: exponential models in context

An investment of 3,000 dollars grows according to $A(t) = 3{,}000e^{0.05t}$ dollars, where t is in years. Find the investment's value after 8 years, rounded to the nearest dollar.

a. 4,200

b. 4,800

c. 4,475 ✓

d. 3,400

$A(8) = 3{,}000e^{0.4} \approx 4{,}475$ dollars.

24 [Foundational] • Target: MCQ — Word problems: exponential models in context

A cup of coffee cools according to $T(t) = 22 + 70e^{-0.08t}$ (°C), where t is in minutes. Find the temperature after 15 minutes, rounded to one decimal place.

a. 38.5 °C

b. 50.0 °C

c. 43.1 °C ✓

d. 22.0 °C

$T(15) = 22 + 70e^{-1.2} \approx 43.1$ °C.

25 [Foundational] • Target: MCQ — Word problems: exponential models in context

A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.

a. 4,000 ✓

b. 2,000

c. 8,000

d. 1,500

$P(9) = 500 \times 2^3 = 4{,}000$.

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26 [Foundational] • Target: MCQ — Word problems: exponential models in context

The value of a car depreciates according to $V(t) = 25{,}000(0.85)^t$ dollars, where t is in years. Find the car's value after 5 years, rounded to the nearest dollar.

a. 11,093 ✓

b. 12,750

c. 9,500

d. 10,625

$V(5) = 25{,}000(0.85)^5 \approx 11{,}093$ dollars.

27 [Foundational] • Target: Solving exponential equations using logarithms

A quantity is modeled by $N(t) = 500e^{-0.1t}$. Find $N(5)$, rounded to one decimal place.

a. 273.0

b. 500

c. 303.3 ✓

d. 333.6

$N(5) = 500e^{-0.5} = 500(0.60653) \approx 303.3$.

28 [Foundational] • Target: Solving logarithmic equations

A quantity is modeled by $N(t) = 5000e^{0.1t}$. Find $N(5)$, rounded to the nearest whole number.

a. 8244 ✓

b. 5000

c. 9068.4

d. 7419.6

$N(5) = 5000e^{0.5} = 5000(1.64872) \approx 8244$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Natural logarithms and e

■■ ■■■■ x ■■ ■■■■■■■■ $\log_3 x + \log_9 x + \log_{27} x = \frac{11}{2}$■ Solve equation with multiple power bases.

a. **27** ✓

b. 9

c. 81

d. 3

■■■■ ■■■■■■ ■■■ 3: $\log_3 x + \frac{1}{2}\log_3 x + \frac{1}{3}\log_3 x = \frac{11}{6}\log_3 x$. ■■■ $\frac{11}{6}\log_3 x = \frac{11}{2} \Rightarrow \log_3 x = 3 \Rightarrow x = 3^3 = 27$.

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30 [Intermediate] • Target: Exponential growth and decay models

■■ ■■■■■ ■■■■ ■■■■■ ■■■■■■■■■ $(\log_3 x)^2 - 5\log_3 x + 6 = 0$■ Product of roots of the logarithmic equation?

a. **243** ✓

b. 81

c. 27

d. 9

$u = \log_3 x \Rightarrow u^2 - 5u + 6 = 0 \Rightarrow u_1 = 2, u_2 = 3$. ■■■■ $x_1 = 3^2 = 9$ ■ $x_2 = 3^3 = 27$. ■■■■ ■■■■■■: $9 \cdot 27 = 243$ (■■ ■■■■■■■ $3^{u_1 + u_2} = 3^5 = 243$).

31 [Intermediate] • Target: Word problems: exponential models in context

■■■■■ ■■■■■■■ $f(x) = ae^{kx}$ ■■■■■■ ■■■■■■. ■■■■ ■■■■ $f(2) = 18$ ■ $f(5) = 486$ ■ ■■■■ ■■ ■■■■■ ■■■■■■ k■ The function $f(x) = ae^{kx}$ represents exponential growth. If $f(2) = 18$ and $f(5) = 486$, what is the growth constant k?

a. $\ln(3)$ ✓

b. $\ln(2)$

c. 3

d. 2

■■■■■■ $f(x) = ae^{kx}$. ■■■■■■■ $f(2) = 18$ ■ ■■■■ $ae^{2k} = 18$ (■■■■■■■■■ 1). ■■■■■■■ $f(5) = 486$ ■ ■■■■ $ae^{5k} = 486$ (■■■■■■■■■ 2). ■■■■■■ ■■■■■■■■■ 2 ■■■■ ■■■■■■■■■ 1■ ■■■■■ ■■■■: $\frac{ae^{5k}}{ae^{2k}} = \frac{486}{18}$ $e^{5k-2k} = 27$ $e^{3k} = 27$ ■■■■■■■ k ■ ■■■■■ ■■■■■■■■■■■■ ■■■■■■■■ ■■■■■ ■■■■■■■■: $\ln(e^{3k}) = \ln(27)$ $3k = \ln(3^3)$ $3k = 3\ln(3)$ $k = \ln(3)$ We have $f(x) = ae^{kx}$. Given $f(2) = 18$, so $ae^{2k} = 18$ (Equation 1). Given $f(5) = 486$, so $ae^{5k} = 486$ (Equation 2). Dividing Equation.

32 [Intermediate] • Target: MCQ — Evaluating exponential expressions

■■ ■■■■■ ■■■■ ■■■■■ ■■■■■■■■■ $\log_2 x - 3\log_x 2 = 2$■ Product of solutions to $\log_2 x - 3\log_x 2 = 2$?

a. **4** ✓

b. 8

c. 2

d. 1

$u = \log_2 x \Rightarrow u - \frac{3}{u} = 2 \Rightarrow u^2 - 2u - 3 = 0 \Rightarrow (u-3)(u+1) = 0$. ■■■■ $u_1 = 3 \Rightarrow x_1 = 8$■ ■ $u_2 = -1 \Rightarrow x_2 = \frac{1}{2}$. ■■■■ ■■■■■■: $8 \cdot \frac{1}{2} = 4$.

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33 [Intermediate] • Target: MCQ — Evaluating logarithms

■■ ■■■■ ■■■■■■■ $\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \cdots \log_{15} 16$■ Telescoping product of logarithms?

a. **4** ✓

b. 8

c. 2

d. 16

■■■■■■■■ ■■■■ ■■■■■ ■■■■■■: $\frac{\log 3}{\log 2} \cdot \frac{\log 4}{\log 3} \cdots \frac{\log 16}{\log 15} = \frac{\log 16}{\log 2} = \log_2 16 = 4.$

34 [Intermediate] • Target: MCQ — Log laws: product and quotient

■■ ■■■■ ■■■■■■■ $\sum_{k=1}^{9} \log\left(\frac{k+1}{k}\right)$■ Evaluate the telescoping logarithmic sum.

a. **1** ✓

b. 0

c. 9

d. 10

■■■■■ ■■■■■■■■■■■■ ■■■■■ ■■■■■■■■ ■■■■ ■■■■■: $\log\left(\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{10}{9}\right) = \log(10) = 1.$

35 [Intermediate] • Target: MCQ — Log laws: the power rule

■■ ■■ ■■■■■■■ ■■■■■■■■ ■■■■■ ■■■■■ ■■■■■■■■ ■■■■■ Which function represents exponential decay?

a. $f(x) = \left(\frac{1}{3}\right)^x$ ✓

b. $f(x) = 3^x$

c. $f(x) = (1.5)^x$

d. $f(x) = 2^x$

■■■■■ ■■■■■■■ ■■■■■■■■■■ ■■■■■■ ■■■■ ■■■■ ■■■■■■■■ $0 < b < 1$. ■■■■ $\frac{1}{3}$ ■■■■ 0 ■ 1.

36 [Intermediate] • Target: MCQ — Expanding and condensing logarithmic expressions

■■■■ ■■■■ $\log_2 x + \log_2 y = 5$ ■ $\log_2\left(\frac{x}{y}\right) = 1$■ ■■■■ ■■■■■ x■ Find x in the logarithmic system.

a. **8** ✓

b. 4

c. 16

d. 2

■■■■ ■■■■■■■■■■■■: $2\log_2 x = 6 \Rightarrow \log_2 x = 3 \Rightarrow x = 2^3 = 8.$

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37 [Intermediate] • Target: MCQ — Solving exponential equations (matching bases)

■■■■ ■■■■■■ $f(x) = a \cdot b^x$ ■■■■ ■■■ ■■■ ■■■ ■■■ $a > 0$ ■: Condition for exponential growth?

a. $b > 1$ ✓

b. $0 < b < 1$

c. $b < 0$

d. $b = 1$

■■■■ ■■■■■■ ■■■■ ■■■ ■■■ ■■■■■ ■■■■ ■■■■■■ $b > 1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Solving exponential equations using logarithms • Cross-Lesson: 1.01 Algebra & Equations

■■■ ■■■ $\log_3(x^2 - 2x + 1) = 2$■ ■■■■■ ■■■ x ■■■■ ■■■■ ■■■ ■■■■■■■■. If $\log_3(x^2 - 2x + 1) = 2$, find the values of x that satisfy this equation.

a. **x = 4 ■■ x = -2** ✓

b. x = 2 ■■ x = -4

c. x = 4 ■■ x = 2

d. x = -2 ■■ x = -4

■■■ ■■■■■■■■ $\log_3(x^2 - 2x + 1) = 2$■ ■■■■■■ ■■■ ■■■■■■ ■■■■■■. ■■■■ ■■ $\log_b a = c$ ■■■■■ $b^c = a$. ■■■■■■■■ $3^2 = x^2 - 2x + 1$. ■■■■ ■■■■■■■■: $9 = x^2 - 2x + 1$ ■■■■ ■■■■■■■■ ■■■ ■■■ ■■■■ ■■■■■■ ■■■■■■ ■■■■■■■: $x^2 - 2x + 1 - 9 = 0$ $x^2 - 2x - 8 = 0$ ■■■ ■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■ ■■ ■■■■■■■■■ ■■■■■■■■ ■■■■■■. ■■■■■■■■■■ ■■■■■ ■■ ■■■■■■ ■■■■■ ■■■■■■■ -8 ■■■■■■■■■■ -2. ■■■■■ ■■■■■■■■ ■■■ -4 ■ 2. ■■■■■■ ■■■■■ ■■■■■■ ■■■■■■■■■ ■■■■■■■■: $(x - 4)(x + 2) = 0$ ■■■ ■■■■■■ ■■■■■: $x - 4 = 0 \Rightarrow x = 4$.

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39 [Challenging] • Target: MCQ — Solving logarithmic equations • Cross-Lesson: 2.01 Functions & Transformations

■■■ ■■■■ $\log_a(x) = 5$ ■ $\log_a(y) = 2$■ ■■■■■ ■■■■■ $\log_a\left(\frac{x^3}{y^2}\right)$. If $\log_a(x) = 5$ and $\log_a(y) = 2$, find the value of $\log_a\left(\frac{x^3}{y^2}\right)$.

a. **23** ✓

b. 17

c. 21

d. 31

■■■■■■ ■■■■■ ■■■■■■■■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■. We use the properties of logarithms to simplify the required expression. ■■■■■: $\log_a\left(\frac{x^3}{y^2}\right)$. We have: $\log_a\left(\frac{x^3}{y^2}\right)$. ■■■■■■■■■ ■■■■■ ■■■■■■■ ■■■■■■■■■■■■: $\log_a(M/N) = \log_a(M) - \log_a(N)$. Using the quotient property of logarithms: $\log_a(M/N) = \log_a(M) - \log_a(N)$. $\log_a\left(\frac{x^3}{y^2}\right) = \log_a(x^3) - \log_a(y^2)$.
$\log_a\left(\frac{x^3}{y^2}\right) = \log_a(x^3) - \log_a(y^2)$. ■.

40 [Challenging] • Target: MCQ — Natural logarithms and e • Cross-Lesson: 3.01 Sequences & Series

■■■ ■■■■■ $f(x) = a \cdot b^x$ ■■■■■ ■■■■■ ■■■■■ $f(1) = 12$ ■ $f(3) = 108$■ ■■■■■■ ■■■■■ a. If $f(x) = a \cdot b^x$ is an exponential function such that $f(1) = 12$ and $f(3) = 108$, find the value of a.

a. 1

b. 2

c. **3** ✓

d. 4

■■■■■■ ■■■■■■■ ■■■■■■■ $f(x) = a \cdot b^x$. We have the exponential function $f(x) = a \cdot b^x$. ■■■■■ ■■ $f(1) = 12$■ ■■■■: Given that $f(1) = 12$, then: $a \cdot b^1 = 12$ (■■■■■■■■■ 1) $a \cdot b^1 = 12$ (Equation 1) ■■■■■ ■■ $f(3) = 108$■ ■■■■: Given that $f(3) = 108$, then: $a \cdot b^3 = 108$ (■■■■■■■■■ 2) $a \cdot b^3 = 108$ (Equation 2) ■■■■■ ■■■■■■■■■ 2 ■■■■ ■■■■■■■■■ 1: Divide Equation 2 by Equation 1: $\frac{a \cdot b^3}{a \cdot b^1} = \frac{108}{12}$ $b^{3-1} = 9$ $b^2 = 9$ ■■■■ ■■ b ■■ ■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■.

41 [Challenging] • Target: MCQ — Exponential growth and decay models • Cross-Lesson: 4.01 Trigonometry

■■ ■■■■■ $\log_3 81$? What is the value of $\log_3 81$?

a. 2

b. 3

c. **4** ✓

d. 9

■■■■■■■ ■■■■■ $\log_3 81$■ ■■■■■ ■■ ■■■■■■■ ■■■■■ ■■■■ ■■■■■ ■■■■ ■■■■■■■■ 3 ■■■■■■■ ■■■■ 81. ■■■■■■■ ■■■■■■ ■■■■ ■■ $3^x = 81$. ■■■■ ■■■■■ ■■ $3^1 = 3$■ $3^2 = 9$■ $3^3 = 27$■ ■ $3^4 = 81$. ■■■■■■ $x = 4$. To find the value of $\log_3 81$, we are looking for the exponent to which the base 3 must be raised to obtain 81. We can write this as $3^x = 81$. We know that $3^1 = 3$, $3^2 = 9$, $3^3 = 27$, and $3^4 = 81$. Therefore, $x = 4$.

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42 [Challenging] • Target: MCQ — Word problems: exponential models in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

■■■ ■■■ $\log_x 64 = 3$■ ■■■ ■■■■ x■ If $\log_x 64 = 3$, find x.

a. **4** ✓

b. 8

c. 2

d. 16

■■■■■■ ■■■■■■■■ ■■■■■■■■■ ■■ Exponentials & Logarithms: $x^3 = 64 = 4^3 \Rightarrow x = 4$. ■■■■■■■■■ ■■■■ ■■■■■■■■ ■■■■■■■■ ■■ 4. Applying Exponentials & Logarithms principles: $x^3 = 64 = 4^3 \Rightarrow x = 4$. Therefore, the correct choice is 4.

43 [Challenging] • Target: Evaluating exponential expressions • Cross-Lesson: 6.01 Limits & Derivatives

■■■ ■■■ $2^x \cdot 4^y = 32$ ■ $3^{x-y} = 9$■ ■■■ ■■■■ x■ Find x in the exponential system.

a. **3** ✓

b. 2

c. 4

d. 1

■■■■■■■■ ■■■■■■■: $2^x \cdot 2^{2y} = 2^5 \Rightarrow x + 2y = 5$. ■■■■■■■■■ ■■■■■■■■: $3^{x-y} = 3^2 \Rightarrow x - y = 2 \Rightarrow x = y + 2$. ■■■■■■■■■: $(y + 2) + 2y = 5 \Rightarrow 3y = 3 \Rightarrow y = 1$■ ■■■■■■■■■ $x = 3$.

44 [Challenging] • Target: Evaluating logarithms • Cross-Lesson: 7.01 Integration & Applications

■■ ■■■■■■ ■■ ■■■■■■■■■■ $\log_2 x > \log_4 (x + 2)$■ Solve $\log_2 x > \log_4 (x + 2)$.

a. $(2, \infty)$ ✓

b. $(0, 2)$

c. $(-1, 2)$

d. $[2, \infty)$

$\log_4 (x+2) = \frac{1}{2}\log_2 (x+2) \Rightarrow \log_2 x > \frac{1}{2}\log_2 (x+2) \Rightarrow \log_2 (x^2) > \log_2 (x+2) \Rightarrow x^2 - x - 2 > 0 \Rightarrow (x-2)(x+1) > 0$. ■■■ ■■ $x > 0$■ ■■■■■ ■■ $(2, \infty)$.

45 [Challenging] • Target: Log laws: product and quotient • Cross-Lesson: 9.01 Statistics & Probability

■■ ■■ ■■■■■■■■ $\log_x 3 + \log_3 x = 2$ ■■■ $x > 0, xe1$■ Solve $\log_x 3 + \log_3 x = 2$.

a. **3** ✓

b. 9

c. 1

d. $\frac{1}{3}$

$\log_x 3 = \frac{1}{\log_3 x}$. ■■■■ $u = \log_3 x$: $\frac{1}{u} + u = 2 \Rightarrow u^2 - 2u + 1 = 0 \Rightarrow (u-1)^2 = 0 \Rightarrow u = 1 \Rightarrow \log_3 x = 1 \Rightarrow x = 3$.

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